CSE 641 Computer Vision Project Report — Week 3

Project Title: Athletic Readiness: Data Generation

Group: Deep Vision

1. Introduction

This week focused on preprocessing and exploratory data analysis (EDA) of the NBA player performance dataset collected during Week 2. The objective was to clean the raw data, correct structural issues, convert feature types, encode categorical variables, analyze statistical properties, and produce a normalized dataset suitable for generative model training such as Variational Autoencoders (VAE) and Generative Adversarial Networks (GAN).

2. Dataset Description

The dataset was scraped from ESPN and contains player statistics from the 2020–2025 NBA seasons. Initially, the dataset contained 1,322 records and 23 columns including player name, position, season information, and 19 performance metrics.

Two categorical features were identified: NAME and POS. The RK column represented a string-based rank and was removed because it was not a meaningful modeling feature. After correction, the dataset contained 22 usable columns.

3. Data Inspection and Cleaning

Initial inspection was performed using df.info() and df.head() to understand the dataset structure and detect inconsistencies. The inspection revealed that the player name column was labeled 'Name' instead of 'NAME', and the RK column contained string values that could interfere with numeric processing. The column was renamed and the RK column removed to ensure consistent data formatting.

4. Handling Missing Values

Missing values were evaluated using df.isnull().sum(). After converting numeric columns with pd.to_numeric(errors='coerce'), rows containing invalid numeric values were checked. No missing values remained, so the dataset retained all 1,322 original records.

5. Data Type Conversion

In the initial notebook only three features (PTS, REB, AST) were converted to numeric types. However, several other performance features were incorrectly stored as object strings. To correct this, all non-categorical columns were converted using pd.to_numeric(errors='coerce'). This ensured accurate statistical calculations and prepared the dataset for machine learning workflows.

6. Categorical Feature Encoding

The POS (player position) column is categorical and cannot be directly used by machine learning models. Therefore, LabelEncoder was applied to create a numeric representation called POS_ENC. This allowed player positions to be incorporated as model input features while preserving the original POS column for interpretability.

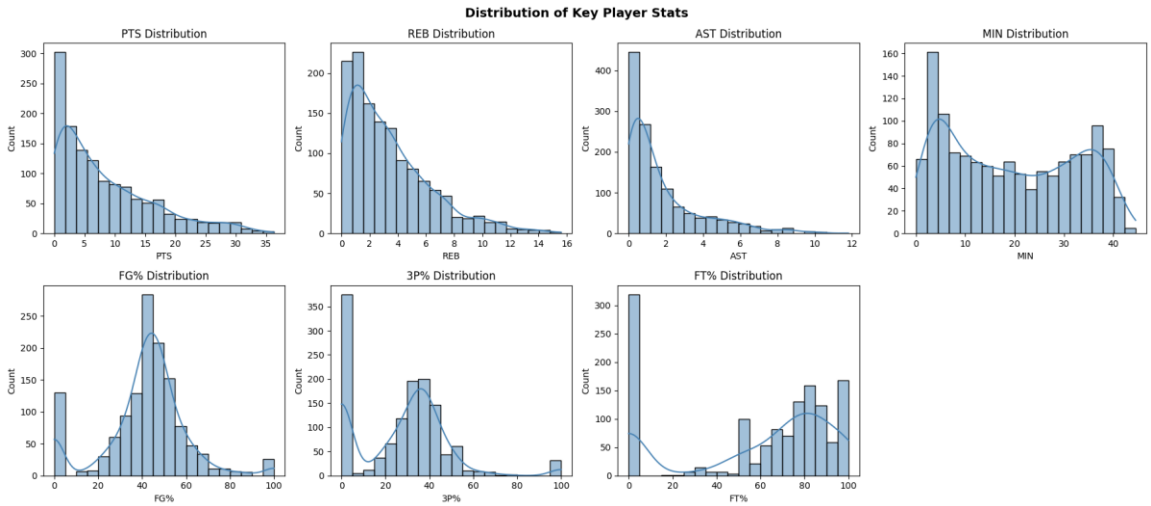
7. Exploratory Data Analysis

Descriptive statistics were computed for all numeric features including mean, standard deviation, minimum, maximum, and median values. A coefficient of variation (CV = std/mean) was also calculated to measure relative variability between player statistics.

The analysis revealed large variation across performance metrics, especially scoring statistics such as points, shot attempts, and assists. Rare events such as double-doubles (DD2) and triple-doubles (TD3) showed extremely high variability because they occur only for a small number of elite players.

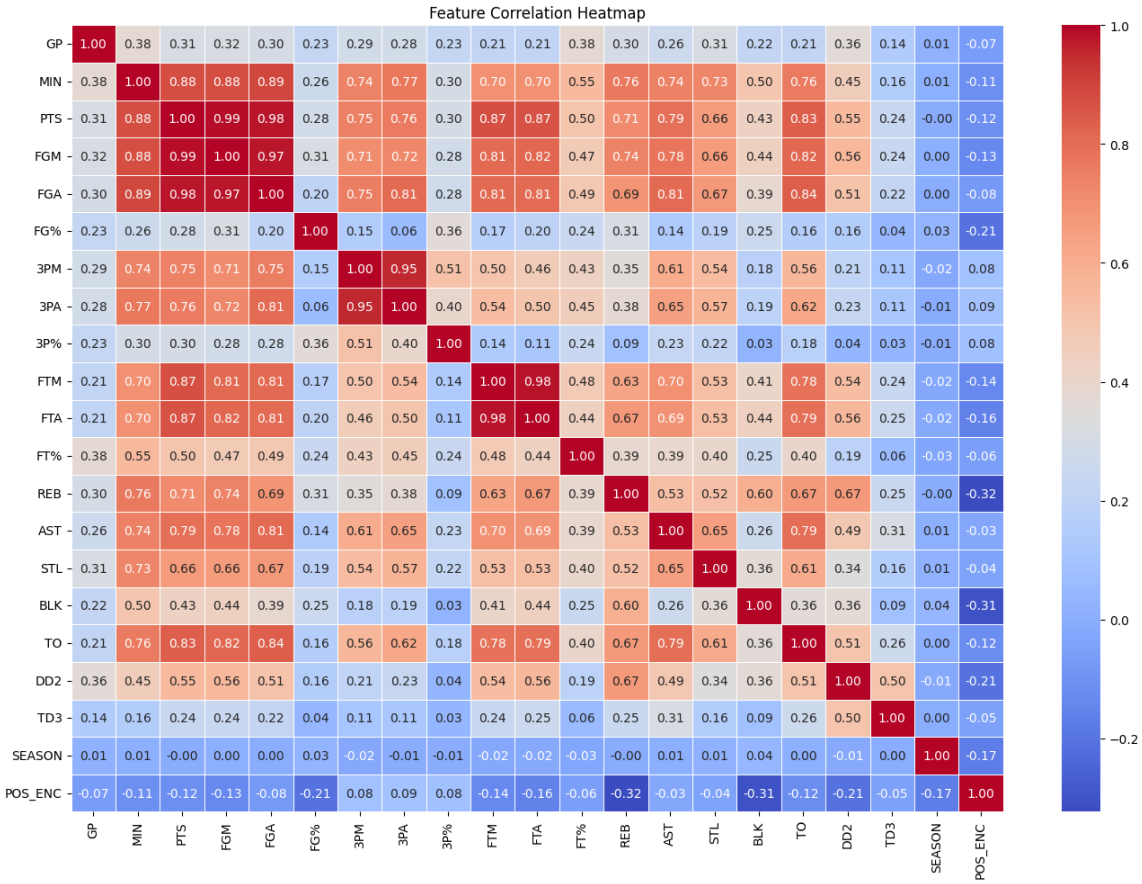
8. Distribution Analysis

Feature distributions were visualized using histograms with kernel density estimation (KDE) curves for key metrics including PTS, REB, AST, MIN, FG%, 3P%, and FT%. Most performance metrics displayed right-skewed distributions. This pattern is typical in professional sports where a small number of elite players produce significantly higher statistics than the majority of players.



9. Correlation Analysis

A Pearson correlation heatmap was generated to identify relationships between variables. The analysis showed extremely strong correlations between points scored (PTS), field goals made (FGM), field goal attempts (FGA), and minutes played (MIN). This indicates that scoring output is largely determined by playing time and shooting volume rather than shooting efficiency.



10. Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method. A record was classified as an outlier if any numeric value fell outside the range Q1 − 1.5×IQR to Q3 + 1.5×IQR.

The analysis identified 586 outlier rows (44.3% of the dataset). After removing these records, the dataset contained 736 rows. This high outlier rate reflects the long statistical tail of elite NBA players whose performance differs significantly from average players.

11. Feature Scaling

After cleaning and outlier removal, StandardScaler was applied to normalize numeric features. Scaling transformed the data to zero mean and unit variance, which improves training stability for neural network models.

Two output datasets were produced:
• nba_clean_dataset.csv – cleaned dataset (736 records)
• nba_scaled_dataset.csv – normalized dataset prepared for model training

12. Key Observations

The EDA process revealed several important patterns. Player statistics exhibit strong right-skewed distributions, scoring is primarily influenced by shot volume and playing time, and efficiency metrics such as shooting percentages show weaker relationships with total points

scored. Additionally, position categories show class imbalance, which may influence model evaluation.

13. Plan for Week 4

In Week 4, the focus will be on selecting the important feature attributes from the processed dataset that represent player performance. After feature selection, the Variational Autoencoder (VAE) architecture will be studied to understand the encoder, decoder, and latent space representation. The objective is to prepare the model architecture and understand how the VAE will be used for synthetic data generation

14. Conclusion

Week 3 successfully completed the data preprocessing and exploratory analysis stage of the project. The dataset was cleaned, structurally corrected, encoded, analyzed, and normalized to create a high-quality dataset suitable for generative modeling. This processed dataset forms the foundation for implementing VAE-based synthetic data generation in Week 4.